

NIKOLAYEVSK-ON-AMUR, Russian Federation

WMO: 313690

Lat: 53.148N

Lon: 140.694E

Elev: 150

StdP: 14.62

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.6	(c) -23.3	(d) -31.6	(e) 0.9	(f) -26.0	(g) -28.1	(h) 1.1	(i) -22.6	(j) 30.2	(k) 8.6	(l) 24.5	(m) 5.2	(n) 2.9	(o) 350	(p) 0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	78.9	67.4	75.5	65.7	72.7	64.2	69.4	76.0	67.4	73.3	65.5	71.1	8.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	99.5	73.1	64.8	92.4	71.0	63.1	86.8	68.9	33.7	76.5	32.1	73.6	30.6	71.2	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	19.0	16.5	-31.3	85.8	4.2	3.8	-34.4	88.6	-36.9	90.8	-39.2	92.9	-42.3	95.7
			-31.6	72.9	4.2	2.9	-34.6	75.0	-37.1	76.7	-39.4	78.3	-42.5	80.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	-6.3	-2.1	11.9	28.2	42.0	55.5	62.2	61.8	51.8	36.2	14.1	-2.5	
	DBStd	26.11	11.16	9.25	9.13	6.73	6.70	6.75	5.67	5.18	6.70	6.94	11.85	11.61	
	HDD50	8521	1746	1458	1181	655	263	24	1	0	59	430	1077	1627	
	HDD65	13008	2211	1878	1646	1105	713	293	121	129	399	894	1527	2092	
	CDD50	1058	0	0	0	0	14	188	379	365	111	1	0	0	
	CDD65	71	0	0	0	0	0	7	34	29	1	0	0	0	
	CDH74	479	0	0	0	0	3	86	200	180	10	0	0	0	
	CDH80	85	0	0	0	0	0	19	43	23	0	0	0	0	
Wind	WSAvg	7.5	6.7	6.3	7.5	8.7	9.3	8.2	7.8	6.5	6.3	7.3	7.4	7.5	
Precipitation	PrecAvg	25.6	1.4	1.2	1.4	1.5	2.0	1.8	2.6	3.4	3.3	3.0	2.2	2.0	
	PrecMax	35.4	4.2	3.9	4.9	4.3	5.2	5.9	6.1	8.7	7.3	8.6	6.7	4.3	
	PrecMin	17.2	0.1	0.1	0.0	0.1	0.0	0.1	0.0	0.3	0.6	0.6	0.2	0.2	
	PrecStd	4.4	1.1	0.9	1.1	0.8	1.1	1.3	1.8	1.9	1.5	1.7	1.2	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.8	26.8	37.5	49.9	69.4	82.5	85.8	83.3	75.4	60.6	41.1	31.9	
		MCWB	27.7	25.1	31.5	40.4	56.1	66.1	72.7	69.2	64.6	50.8	37.2	31.2	
	2%	DB	24.7	21.3	31.8	44.2	63.0	75.4	78.8	78.6	71.3	54.3	35.8	28.6	
		MCWB	23.7	19.7	28.5	37.3	52.3	63.5	67.7	67.9	61.6	47.8	33.9	27.6	
	5%	DB	17.5	16.5	29.1	40.8	58.4	71.6	75.2	75.5	67.5	50.3	33.4	23.7	
		MCWB	16.5	15.1	26.7	35.1	49.6	61.4	66.0	66.3	59.7	45.4	31.7	22.9	
	10%	DB	10.2	12.2	26.6	37.6	54.0	68.2	72.0	72.4	63.5	46.8	30.9	15.0	
		MCWB	9.4	11.1	24.1	33.3	47.1	59.7	64.7	64.9	57.5	43.0	29.2	14.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.9	25.4	32.1	41.4	57.2	68.4	72.9	72.4	66.7	52.7	37.7	31.1
			MCDB	28.7	26.6	35.8	47.6	68.1	78.2	82.4	79.2	72.7	57.4	40.2	31.4
2%		WB	23.7	19.7	29.1	37.9	53.1	65.1	69.5	69.7	63.0	49.2	34.1	27.6	
		MCDB	24.7	21.0	31.3	43.2	61.1	73.3	76.5	75.7	69.0	53.2	35.1	28.5	
5%		WB	16.6	14.9	26.9	35.6	50.6	62.7	67.4	67.8	60.6	46.0	31.8	22.9	
		MCDB	17.5	16.3	29.1	39.4	56.8	69.8	73.3	73.2	65.6	49.2	32.8	23.9	
10%		WB	9.4	10.9	24.2	33.7	48.0	60.5	65.6	65.9	58.3	43.5	29.1	14.0	
		MCDB	10.0	12.1	26.4	36.9	53.3	67.1	70.8	71.3	63.1	46.3	30.7	14.6	
Mean Daily Temperature Range		5% DB	MDBR	13.4	17.1	19.3	15.5	15.8	18.0	15.0	16.3	18.3	14.5	13.7	12.5
			MCDBR	16.2	17.2	19.1	20.1	25.1	25.4	22.3	23.0	24.3	21.3	13.2	12.6
	MCWBR		16.0	15.6	15.9	14.7	15.8	15.2	13.1	13.6	15.4	15.0	11.5	12.4	
	5% WB	MCDBR	15.9	16.1	17.5	17.7	23.3	22.6	18.4	19.6	20.9	18.2	11.6	12.4	
		MCWBR	15.7	14.8	14.7	13.4	15.3	14.2	11.6	12.6	14.2	13.9	10.6	12.4	
Clear-Sky Solar Irradiance	taub	0.286	0.312	0.384	0.493	0.555	0.522	0.523	0.457	0.391	0.393	0.344	0.299		
	taud	2.000	1.964	1.807	1.642	1.715	1.932	1.981	2.184	2.311	2.105	2.014	1.960		
	Ebn at Noon	205	239	242	229	224	235	232	242	246	208	176	168		
	Edh at Noon	25	36	52	71	69	56	53	41	32	31	25	22		
All-Sky Solar Radiation	RadAvg	334	683	1111	1435	1552	1720	1530	1301	1012	571	309	227		
	RadStd	39	54	108	155	171	180	188	121	89	52	37	42		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-254	N/A	N/A
(47) Regional (3 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-254	N/A	N/A

Nomenclature: See separate page